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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VII Semester B. E. Examinations March -2022

Computer Science and Engineering

PARALLEL ARCHITECTURE AND DISTRIBUTED PROGRAMMING

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Identify the relationship between Warps, thread blocks and CUDA cores.	01
	1.2	With a suitable example justify that name dependence is not a true data dependence.	02
	1.3	Identify the three different effects that limit the gains from loop unrolling.	02
	1.4	What are <i>correlating Branch Predictors</i> ?	02
	1.5	Define Imprecise exceptions.	01
	1.6	Enumerate the use of strip mining.	02
	1.7	What are loop carried dependences?	02
	1.8	Use the GCD test to determine whether dependences exist in the following loop: $\begin{aligned} & \text{for } (i = 0; i < 100; i = i + 1) \\ & \quad \{ \\ & \quad \quad X[2 * i + 3] = X[2 * i] * 5.0; \\ & \quad \} \end{aligned}$	02
	1.9	Compare write invalidate and write update snooping based cache coherence protocol.	02
	1.10	Give the prototype of MPI_Comm_Split and MPI_Cart_Sub routines used for partitioning the groups and communicators.	02
	1.11	Identify the issues addressed by block cyclic distribution.	02

PART-B

2	a	Analyze the differences between x86 and MIPS processors by comparing their dimensions of Instruction Set Architectures.	08
	b	Illustrate the basic structure of a FP unit using Tomasulo's algorithm extended to handle speculation.	08
3	a	Analyze write invalidate snooping cache coherence protocol for a write back cache showing the states and state transitions for each block in the cache.	08

4	b	Compare shared memory and distributed memory programming models.	04
	c	Examine LoadLinked and StoreConditional instructions with an example.	04
		OR	
	a	Illustrate directory based cache protocol indicating the actions to which each individual cache responds with a supporting state transition diagram.	08
	b	Classify the various Relaxed Consistency models.	04
	c	Analyze the following basic hardware primitives used for synchronizaiton: i. TestAndSet ii. FetchAnd Increment iii. Loadlinked and StoreConditional.	04
5	a	Consider the DAXPY loop that forms the inner loop of the Linkpack benchmark. $Y = a \times X + Y$, X and Y are vectors, initially resident in memory, and a is a scalar. Show the code for MIPS and VMIPS for this loop. Assume that the starting address of X and Y are in R_x and R_y respectively. Indicate the performance gain obtained using VMIPS.	07
	b	Analyze the major innovations introduced by Fermi to bring GPUs much closer to mainstream system processors.	09
		OR	
6	a	The following loop has multiple types of dependences. Find all the true dependences, output dependences and anti-dependences and eliminate the output dependences and anti-dependences by renaming $\text{for}(i = 0; i < 100; i = i + 1)$ $\{$ $Y[i] = X[i] / c; /* S1 */$ $X[i] = X[i] + c; /* S2 */$ $Z[i] = Y[i] + c; /* S3 */$ $Y[i] = c - Y[i]; /* S4 */$ $\}$	08
	b	Illustrate the typical processing flow of a CUDA program highlighting all the important components. Write a CUDA C program to compute the product of two matrices.	08
7	a	Write a MPI program that sort a sequence of n elements using simple sorting algorithm.	08
	b	Identify the characteristics of the tasks that have a high influence on sustainability of a mapping scheme.	08
8	a	Explain device management and kernel launch in OpenCL	08
	b	With a supporting diagram, illustrate OpenCL data parallelism model.	08